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PRACTICAL APPLICATION

# Qualitative In-trial Interviews: Methods, Challenges, and Best Practice

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## Abstract

Qualitative in-trial interviews with clinical trial participants are a means of providing rich, in-depth patient experience data to supplement and complement data captured by clinical outcome assessments and other clinical trial efficacy endpoints. Such in-trial interview data can be used to build understanding of disease and treatment experiences, evaluate content validity of clinical outcome assessments, aid interpretation of scores and meaningful changes, inform trial design feasibility and operational considerations, and provide supportive evidence regarding safety, efficacy, and effectiveness. Despite the rapid growth of in-trial interviews as part of clinical development programs in the pharmaceutical industry in recent years, published guidelines regarding the methods, conduct, and implementation of in-trial interviews are scarce. Drawing on published examples and the authors' experiences of conducting in-trial interview studies, this article provides an overview of best practice methods for implementing this methodology (including considerations for study design, sample size, interview conduct, and analysis) and the value of in-trial interview data to answer specific research questions. Operational and logistical considerations are outlined, including recommendations for country selection, site selection, training and communication, adverse event safety reporting, and data management and handling. Well-designed and carefully implemented in-trial interviews can lead to generation of insightful patient experience data that are truly of value to inform regulators, health technology agencies, clinicians, patients, and caregivers about product attributes and the impact of diseases and treatments on patients' lives.

**Key Points for Decision-Makers**

Qualitative in-trial interviews with clinical trial participants can be used across all clinical trial phases to collect patient experience data that answer a range of research questions. These could include research questions relating to the patient's experience of their condition, completing assessments during the trial, and using the trial treatment.

The value of the findings generated is increasingly recognized by a range of different stakeholders, including patients, caregivers, clinicians, regulators, and health technology agencies.

This article addresses the limited guidance currently available regarding best practice for conducting these studies, including the methods, conduct, and implementation of in-trial interviews.

Extended author information available on the last page of the article

## 1 Introduction

Qualitative in-trial interviews are interviews conducted with clinical trial participants and can be used to collect patient experience data (PED). In-trial interviews can be conducted for various purposes to meet the needs of different stakeholders, including patients, caregivers, clinicians, regulators, and health technology assessment (HTA) bodies. Common objectives for conducting in-trial interviews include increasing understanding of the patient experience of a disease and treatment, exploring changes in symptoms or functioning experienced throughout the clinical trial and whether changes experienced are meaningful and important, evaluating content validity and usability of clinical outcome assessments (COAs), aiding interpretation of clinically meaningful change thresholds of COA scores, and obtaining feedback on clinical trial participation and study procedures [1–3].

Increasingly, PED collected through qualitative in-trial interviews are used to supplement traditional clinical trial endpoints [4]. Use of this methodology has been, in part, stimulated by recommendations from regulators such as the US Food and Drug Administration (FDA) and as advocated in the patient-focused drug development (PFDD) guidance

series [5–9]. Over the last decade, discussion of the methodology has been growing within the literature, and more studies have been published across different clinical trial phases and disease areas [4, 10, 11].

Despite the rapid growth of in-trial interviews, published guidelines regarding optimal methods, implementation, and analysis remain limited. This article aims to provide an overview of best practice methods, operational considerations for implementing in-trial interviews, and opportunities to generate data that have the potential to support regulatory decision-making and market access, drawing on published examples and the authors' experiences of conducting these studies.

## 2 In-Trial Interview Methods

Various terms are used to refer to qualitative interviews conducted with clinical trial participants. The most used term is "exit interviews", which refers to interviews conducted at or around the final efficacy treatment visit of a clinical trial [4, 10, 12–28]. The FDA uses the terms screening and exit interviews in PFDD guidance [5, 6]. In practice, it is rare for only screening interviews to be conducted. The term "in-trial" is typically used when interviews are conducted at more than one timepoint within a clinical trial, for example at screening and exit, or sometimes as a "catch-all" term that could refer to interviews conducted at screening, midpoint, and/or exit. "Embedded interviews" is a term that is sometimes used [15, 29–31] and is perhaps more appropriate when the operational set up of the study is such that the interviews are incorporated into the clinical trial protocol (embedded) rather than being detailed in a separate protocol to the clinical trial (standalone). For this article, the term "in-trial interviews" has been used as an umbrella term to refer to the methodology.

In-trial interviews are qualitative, semi-structured interviews that commonly employ similar methodology to that used in standalone qualitative studies, such as concept elicitation (CE) and cognitive debriefing (CD) techniques. A combination of open and closed questions is typically used to elicit unbiased participant perspectives and ensure that all topics are covered [14]. A range of participants (e.g., patients, caregivers, and clinicians, among others) can be interviewed, depending on the interview objectives, population, and condition. Interview durations are typically 30–60 min but may vary according to the research objectives, and 90-min interviews may be warranted if the interviews have multiple objectives.

### 2.1 Research Objectives

In-trial interviews can address various research objectives. Although most objectives can be explored at any trial phase, some may be more relevant or important to explore in earlier phases (Fig. 1). Evidence generation may still be required during a pivotal trial, where it may not have been possible to conduct a standalone qualitative study [6]. In the authors' experience and consistent with a recent literature review [4], in-trial interviews are most frequently conducted within phase III trials. However, there is great value in conducting them in phase I/II trials [32].

#### 2.1.1 Understanding Disease Experience to Inform Clinical Outcome Assessment (COA) Measurement Strategy and Supplement and Aid Interpretation of COA Endpoints

In-trial interviews can increase understanding of disease experience, including symptoms, functioning, and health-related quality of life (HRQoL) impacts and how these change with treatment [12, 22, 23, 25, 32–34]. This can include in-depth exploration of the symptoms experienced at the start of the trial (i.e., descriptions, severity, frequency, duration, and variability), followed by if and how symptoms may have changed. When explored in phase I/II trials, findings can identify and confirm meaningful aspects of health to patients and concepts of interest important to assess in a measurement strategy, informing selection and/or refinement of COAs. In pivotal trials, these insights may be most often used to contextualize and support interpretation of other trial endpoints. In rare diseases, qualitative evidence generated in any trial phase may help compensate for sample size limitations in analyses of primary and secondary efficacy endpoints. These topics can be explored through a CE approach to identify concepts relevant to the patient experience reported spontaneously by participants.

#### 2.1.2 Content Validation of COAs

If content validity evidence has not been generated for a COA (i.e., due to lack of time, budget constraints, or access to patient populations), this evidence can be obtained through in-trial interviews [35]. A CD approach is adopted whereby participants describe their thoughts, for example, using "think aloud" or through being asked detailed questions to assess understanding and relevance for each instruction, item, response scale, and recall period [36–38]. Evidence can be generated for all COAs, including performance-reported outcomes. For relevant COAs and digital health technologies, questioning may focus on concept relevance, concept importance, and (where relevant) task comprehension and consistency [39]. CD interviews are typically performed separately from clinical trials, but a benefit of performing them as part of a trial is that participants can

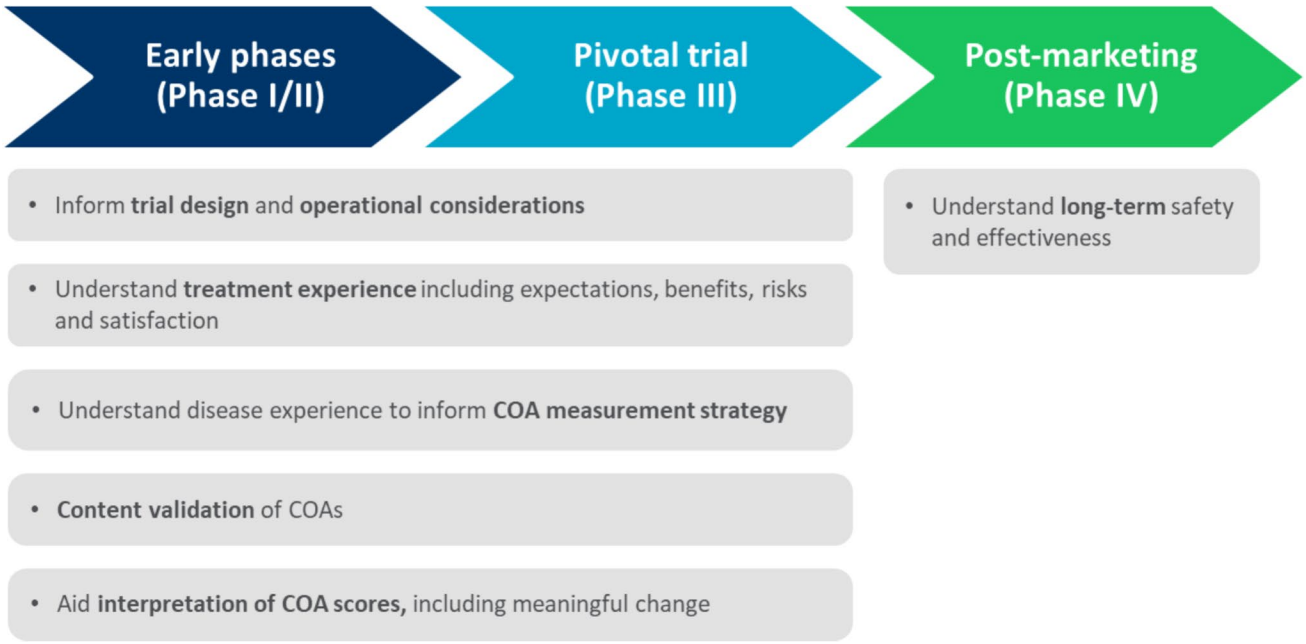



**Fig. 1** Overview of research objectives for in-trial interviews across clinical trial phases. *COA* clinical outcome assessment

reflect upon their experience of completing a COA throughout the trial [11, 24, 26, 40]. However, the timing of CD activities as part of an in-trial interview study must be carefully considered to avoid any potential learning effects or biasing participants' responses to any COAs completed during the clinical trial. This consideration is also noted by the FDA as part of their PFDD series, where in-trial interviews are recommended to be conducted at either screening or exit to avoid compromising data integrity [6] for the main part of the clinical trial (i.e., the period of randomization that will form the basis of the primary efficacy assessments). Further, assessing the content validity of COAs as part of in-trial interviews, particularly as part of later-phase trials, may risk uncovering unfavorable evidence at a time when it may be too late to modify the COA or otherwise adapt the COA strategy. If more than minor modifications to a COA are implemented based on the results, content validity and psychometric validity may need to be re-established for the modified COA. Therefore, where practically possible, it is preferable to obtain content validity evidence in a standalone qualitative study or before a pivotal trial. If it is necessary for it to be performed as part of a pivotal trial, arguably, interviewing participants at exit following completion of trial related procedures and treatment is the best timing to avoid the risk of the interviews influencing responses to the COAs during the trial. It would also be preferable to interview the first participants reaching trial exit, in case updates to the COAs or changes to trial processes are recommended for the remaining participants.

#### 2.1.3 Aid Interpretation of COA Scores Including Meaningful Change

The most common and valuable research objective for in-trial interviews is to explore meaningful change, to understand whether participants perceive changes following treatment as meaningful or important. Data generated using a qualitative approach can inform what level of change on a COA or anchor may be meaningful and/or can provide additional context for meaningful within-patient change thresholds derived from anchor-based analyses, especially if the confidence in thresholds is limited because of small samples.

Qualitative approaches to exploring meaningful change outside a clinical trial have been described in the literature [41, 42]. A similar approach can be used to explore meaningful change during in-trial interviews [41, 42], asking participants to describe the importance of perceived changes experienced in symptoms, functioning, and/or HRQoL impacts throughout the trial [12, 22, 24, 25, 32, 43]. These qualitative insights about key concepts can then be mapped to reported changes in COA or anchor scores [16, 33, 43, 44]. For any participants who do not report changes, meaningful change can be discussed hypothetically to understand what an important change in symptoms, functioning, and/or HRQoL impacts would be.

Alternatively, actual change in COA and/or anchor scores reported during the trial can be shared with participants during an exit interview, and they can then be asked why they selected specific responses, what qualitative changes in

symptoms and functioning experienced are associated with the selected COA or anchor score changes, and whether those changes were meaningful or important. However, this approach does clearly carry additional risks, given that the participant's trial data must be accessed and shared. At a minimum, it requires careful planning to ensure the interviewer can access the COA and/or anchor scores ahead of the interview without any risk of unblinding or undermining trial data integrity (e.g., there is potential risk of misalignment between participant recollections/qualitative comments on their experiences and their actual COA and/or anchor scores). For these reasons, it is recommended that this approach should only be considered when interviews are performed at "exit" or at least after the primary endpoint timepoint. Another approach is to ask participants to respond to "global impression of change" items during the interview to provide feedback on what level of change would be meaningful to them using pre-defined response options [17, 26, 34, 44, 45]. This approach could be considered to assess additional concepts or different perspectives (i.e., patient, caregiver, clinician) using global impression items to complement other anchors or to administer items to the interview sample in the absence of any anchors in the main trial protocol.

#### 2.1.4 Understanding Treatment Experience

In-trial interviews can support understanding of treatment experience by exploring improvements, maintenance, or worsening of disease symptoms, functioning, and HRQoL impacts over the trial [12, 19, 27, 28, 46]. Expectations of and satisfaction with experienced treatment changes and side effects/tolerability, and interest in using the treatment again in the future, can also be explored [19, 28].

#### 2.1.5 Inform Trial Design and Operations

In-trial interviews conducted within early-phase trials can obtain feedback on trial participation, including enrollment/consent process, decisions to participate, support received from site staff, and feedback on study procedures [25, 28]. This is commonly a secondary objective. Findings can inform future trial procedures, minimizing participant burden and ensuring study feasibility [28, 32]. Conducting interviews for this purpose can be particularly valuable for novel or complex trial designs, and CE methods can be used.

#### 2.1.6 Long-Term Safety and Effectiveness

Interviews may be used in later-phase trials (i.e., phase IV) for post-marketing activities to evaluate and monitor the long-term safety and efficacy of products in broader, more "real-world" settings. Feasibility of treatment administration in the real world can also be explored, including preferences regarding mode and frequency of administration and identification of potential logistical and practical considerations for administering treatment at sites (i.e., equipment availability, staff training, frequency of visits) [2].

### 2.2 Embedded In-trial or a Separate Study

Interviews can be embedded into the trial protocol or conducted as an operationally separate, standalone study (i.e., separate study protocol and consent) with participants recruited from the clinical trial study sample. Incorporation into the trial usually maximizes efficiencies, minimizes timelines and costs, and avoids the need for separate ethics approvals, informed consent forms, and site contracts. Additionally, clinical study data can be shared with the agency conducting the in-trial interview study to avoid duplicating data collection. A brief description of the in-trial interviews can be included within the trial protocol. However, a streamlined interview protocol outlining the interview study methods and procedures and a top-line interview guide can be produced and may be requested by regulators and included as an appendix to the trial protocol. Sometimes the decision to add in-trial interviews is made after the trial protocol is finalized or the trial has commenced. Interviews can still be added through a protocol amendment, but it may be more efficient or feasible to conduct a standalone study with a separate protocol, and clinical teams are sometimes reluctant to risk impacting study timelines or data collection by amending the main trial protocol. Participation can either be an activity for all trial participants or an optional activity. Although this will need to be explicitly specified, it can be specified in the informed consent form relating to trial participation or a standalone informed consent form.

### 2.3 Sample Size

In-trial interviews are typically conducted with a subset of the trial sample. A recent review showed that various sample sizes are employed ($n$=10–1993; 0.06–92.00% of trial participants) [4]. The optimal sample size depends on the trial design (e.g., randomized, double-blind trial vs. open label), homogeneity of the population, and research objectives. Depending on the research objectives, sample size selection can be based on concept saturation (defined as the point at which all concepts of importance to participants have been elicited) [47]. In a relatively homogeneous population, concept saturation can be achieved in 10–25 interviews [48–50]. Previous research indicates that 97% of disease-related concepts are elicited within 20 interviews and 99% of concepts within 25 interviews [51]. However, conceptual saturation is not applicable when evaluating content validity of COAs, exploration of meaningful change, evaluation

of treatment experience or feedback on trial participation. For CD interviewing, a base sample size of 12–15 patients is typically adequate to assess understanding and relevance of the content of a measure [52]. Sample sizes should be increased when necessary to ensure adequate representation of demographic and clinical characteristics of the target population (e.g., age groups, subtypes of a condition, education levels that may relate to reading and comprehension abilities).

For randomized, double-blind trials, where the aim is to explore treatment experiences, the sample size should be large enough to expect sufficient representation across treatment arms, recognizing that this cannot be controlled for. In rare diseases where the trial sample size can be small, interviewing the entire trial sample may be feasible, and interview findings may assume greater importance, to help interpret trial endpoints.

## 2.4 Timing, Scheduling, and Conduct

Interviews have traditionally been conducted when participants exit the efficacy phase of a clinical trial or at a key timepoint to align with evaluation of the primary endpoint [53]. Arguably, most insights can be gained regarding treatment experience at exit, minimizing the risk of impacting the integrity of the trial/primary endpoint [6]. However, interviews conducted at screening can provide an initial understanding of disease experiences before receiving any treatment, and these can then be compared against symptoms reported following treatment. Pre-trial disease experience can be explored at exit, but this does involve risks of recall bias, depending on the trial duration, participant type (e.g., patients with cognitive impairments), or disease.

As trial sites can be dispersed geographically and enrollment can stretch over many months, it is optimal to conduct interviews remotely via either telephone or videocall/teleconference software, which has limited impact on data quality [54–58]. Interviews can be scheduled to take place during a study visit (assuming the availability of a private room with appropriate technology) or at home, within a certain timeframe of the study visit (e.g., 7–14 days post-trial visit) to minimize recall bias. It may be more convenient to conduct interviews from the participant's home following the site visit to minimize the risk of social desirability bias or participant fatigue. Interviewers and site staff should work closely to facilitate efficient scheduling of interviews at a date and time convenient for participants, for example, by using an online calendar showing interviewer availability. Site staff can provide participants with dial-in details or a link for an online meeting for their interview to ensure that no personally identifiable data are shared.

In-trial interviews are best conducted by trained qualitative interviewers rather than site staff. Qualitative interviewers are more likely to avoid leading questions or bias and to produce high-quality data [1, 6].

## 2.5 Qualitative Analysis

In-trial interviews can be analyzed using thematic analysis methods [59–62]. Thematic analysis involves identifying and coding patterns, or themes, across interviews; as a theory-free approach, it provides a rich, detailed, and complex synthesis of data to meet a specific aim. Additionally, qualitative analysis can be used to apply prespecified dichotomous codes to capture participant responses regarding the concepts in question, for example, whether an item is understood as intended.

Interim analysis can be performed following completion of a proportion of the interviews, facilitating early sharing of the findings and to support early decision-making. Full analysis can then be conducted once all interviews are complete. Following database lock, treatment allocation data can be shared and analyzed unblinded to explore any treatment group differences, depending on the research objective. Access to demographic and clinical data allow for subgroup differences to be explored and comparisons made between the in-trial interview and full trial samples to evaluate representativeness [63].

## 2.6 Operational Considerations

### 2.6.1 Country Selection

Selection of countries may be driven by disease prevalence and to satisfy regulatory and HTA bodies requirements in key markets by generating country-specific data while balancing costs and timelines (i.e., costs will be reduced if translation is not required). If the FDA has advised that in-trial interviews be conducted, priority may be placed on interviewing a proportion of patients in the USA. However, PED are also of increasing interest to the European Medicines Agency (EMA) [64], and there are examples where HTA bodies such as the Canadian Agency for Drugs and Technologies in Health [65–67], the All Wales Medicines Strategy Group (AWMSG) [11, 45], and the French Haute Autorité de Santé (HAS) have reviewed PED as part of reviews [68]. Although conducting in-trial interviews only in English-speaking countries may reduce costs, it could introduce bias by not representing the experience of the whole trial sample.

### 2.6.2 Site Selection, Training, and Communication

If a subset of the clinical trial sample will be involved in the in-trial interview study, an appropriate number of sites will need to be selected to support recruitment. How engaged,

efficient, and responsive sites are often depends on capacity and resource and can substantially impact recruitment success. If the interviews will be conducted as a standalone study, identifying sites that will accept centralized ethics approval can help minimize costs and timelines. If the study has commenced at the time of site selection, the sponsor and clinical research organization can advise on higher enrolling sites.

Once sites are selected, key contacts should be identified and fully briefed with in-depth training and support materials. Additional training can be conducted throughout the trial, depending on trial duration, to ensure alignment on responsibilities and requirements. Clear communication channels should be established between all parties. If the sponsor receives recruitment updates from sites or the clinical research organization, sharing updates with the agency minimizes site burden and duplication of communication. An overview of site and agency roles and responsibilities is provided in Fig. 2.

#### 2.6.3 Adverse Event Reporting

Adverse events (AEs) can be reported by participants during in-trial interview studies, particularly if the interviews explore treatment experience. AE-reporting procedures should be consistent with clinical trial AE-reporting requirements, and the agency should complete sponsor-specific AE training at study commencement. This may need to be renewed each year, depending on the trial duration. To ensure AEs are not double counted (i.e., reported by the agency and the site separately), any AEs identified during in-trial interviews can be reported to the site for reconciliation, within either 24 h or 1 business day of becoming aware of the AE. A reconciliation form can be submitted to the sponsor, providing a summary of all AEs identified during the in-trial interviews and the reconciliation outcome. Generally, most AEs elicited during interviews have already been reported during the clinical trial, so in-trial interviews do not result in a significant increase in AEs reported.

#### 2.6.4 Compliance

Qualitative in-trial interview studies, whether embedded in the trial or standalone, should follow good clinical practice and data protection regulations, including Health Insurance Portability and Accountability Act guidelines and the European General Data Protection Regulation for the security and privacy of health data. Pseudonymized participant data may be shared between parties to facilitate the conduct of the study, so it is important to ensure data-sharing processes are agreed between parties who require access to trial data to ensure transfer of data securely, using approved systems, and at key points (i.e., during the study and following database lock). A data management plan, a risk-based document, can be developed to document end-to-end data processing and the mechanisms to protect and manage data at different stages and in different locations. A data flow diagram can visualize the secure management and transfer of data across data systems and parties and may be included in the data management plan. The need for such documents varies by sponsor, and they are not required for all in-trial interview studies.

## 3 Discussion and Conclusion

Qualitative interviews with clinical trial participants can provide rich, in-depth insights into the patient experience beyond data captured by traditional COA endpoints. The uptick in application of this methodology is evidenced by the increasing number of publications in the field, and the trend is expected to continue to increase as more studies are completed and data published.

The value of in-trial interviews is increasingly recognized in the pharmaceutical industry, and the FDA is more often requesting the methodology to address evidence gaps and provide supplementary evidence to contextualize and facilitate interpretation of trial data [6, 69]. While in-trial interview data may rarely be looked at in isolation, they can contribute toward the totality of evidence regarding the patient experience alongside more traditional trial endpoints. Findings from in-trial interviews have been used in FDA review documents to highlight the most bothersome symptoms, confirm relevance of concepts, and inform meaningful change thresholds. For example, qualitative exit interviews were conducted in a subset of phase II and III study participants for telotristat ethyl in carcinoid syndrome to understand the patient experience of receiving treatment and explore meaningfulness of changes in symptoms [34, 70]. FDA review staff confirmed that baseline characteristics of the substudy participants were comparable to those of the overall trial population and outlined that the findings confirmed the most salient symptom and aspect of symptom to improve, the significant burden that participants experience, that participants receiving treatment reported being satisfied, and that improvements in these symptoms may have resulted in improved HRQoL. Global impression items were completed by substudy participants to provide an assessment of adequate relief of carcinoid syndrome symptoms, which was accepted as a standalone anchor. Exit interview findings and several anchors were used to inform a meaningful change threshold range for the primary endpoint [71]. More recently, qualitative in-trial interviews were conducted in a phase II/III open-label trial in Chaple disease to explore

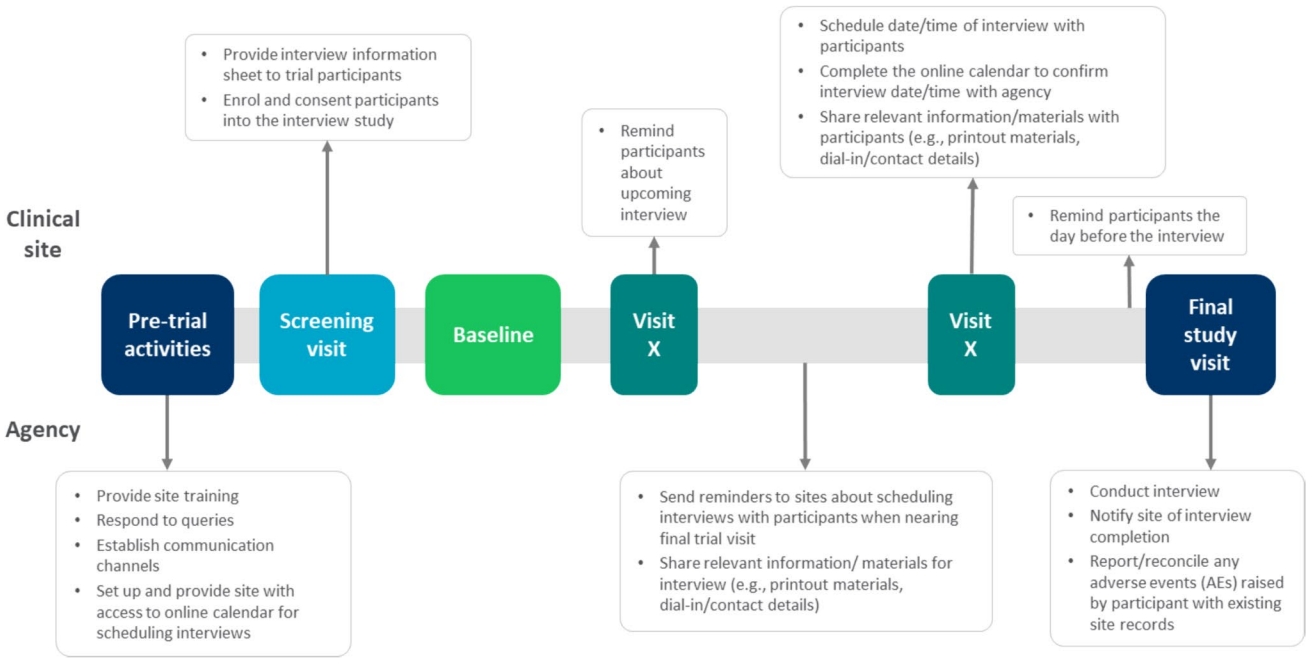



**Fig. 2** Roadmap of roles and responsibilities between sites and vendor for in-trial interviews conducted upon trial exit. *AEs* adverse events

signs/symptoms and impacts of treatment on disease. The FDA acknowledged that the findings confirmed the most bothersome symptoms and demonstrated that participants experienced complete resolution of their core signs/symptoms, which had positive impacts on participants'' daily lives. However, the open-label study design was mentioned as a limitation to the ability to draw conclusions from the study [33].

Beyond the FDA, the value of in-trial interview data to other regulators and HTA bodies is less clear. The EMA public assessment report for telotristat ethyl mentioned qualitative exit interviews with participants receiving treatment to evaluate the "relevance and clinical meaningfulness of symptom improvements", level of change in symptoms experienced, and satisfaction with treatment. To the authors' knowledge, this is the only example where the EMA have publicly mentioned in-trial interview data [72]. Some HTA bodies have mentioned in-trial interview data in their reports, such as the UK National Institute for Health and Care Excellence, HAS, the AWMSG, and the Scottish Medicines Consortium [11, 18, 45, 73, 74]. These individual use cases provide some insights into the potential thinking of some HTA bodies. For example, the UK National Institute for Health and Care Excellence described the two qualitative interview studies for telotristat ethyl, provided a quality criteria rating for each study, and outlined that the results suggest that "patients do report improvements in their carcinoid syndrome symptoms as a result of taking telotristat" [73]. However, they highlighted that results should be interpreted cautiously because of the small sample size and that participants may have received different doses [73]. Similarly, the HAS noted that no conclusions could be drawn from the qualitative exit interview study because of the small sample, which may not be representative [68]. The AWMSG and the Scottish Medicines Consortium mentioned the findings to support reimbursement decision-making, and the AWMSG highlighted the added value of data for economic evaluation and complementing cost-utility measures [11, 45, 74].

The authors are unaware of any examples of in-trial interview data being included in review documents by the Japanese Pharmaceuticals and Medical Devices Agency, the German Federal Joint Committee (Gemeinsamer Bundesausschuss), or other European HTA bodies. This may be because products have not been submitted for approval or reimbursement or have not been approved or reimbursed in these countries. Additionally, in-trial interview data may not have been submitted, or documents may not be available publicly in English.

Despite the growing use of in-trial interviews and case studies where data have been submitted to regulators and HTA bodies, the extent to which in-trial interview data have typically informed decision-making remains unclear. This may be because it is difficult to isolate the individual contribution of in-trial interview data from the totality of evidence submitted or because there are not yet many examples where in-trial interview data have substantially influenced approvals. However, the same is true for PED more broadly, and more guidance and harmonization on what regulators

and HTA bodies value beyond COAs would be welcomed. Sponsors are encouraged, where feasible, to submit in-trial interview data as part of regulatory and HTA submissions to provide supplementary evidence to contextualize and improve understanding of treatment benefits and meaningful improvements in symptoms.

This article has some limitations. Our intention was to summarize best practice for the design, planning, conduct, and analysis of in-trial interview studies to provide an overview of different approaches to and ways of operationalizing studies. However, it is possible that other scenarios are not mentioned here. Additionally, this article did not cover reporting data from in-trial interview studies, and future research could focus on this and optimal ways to present the detailed qualitative data that can emerge from such interviews.

Qualitative in-trial interview studies can generate rich, in-depth PED following treatment, aid exploration of meaningful outcomes, and facilitate interpretation of clinical trial endpoints. Although the studies can be complex to plan and operationalize, data generated can be invaluable to support both communication of treatment experiences to patients, caregivers, and clinicians and the evaluation of risks versus benefits by regulators and HTA bodies [34, 75].

## Declarations

**Funding** No funding was received for conducting this study.

**Conflicts of Interest** NW was an employee of Adelphi Values at the time of authoring this article. NW is currently an employee and shareholder of UCB Biopharma, Slough, United Kingdom. CH, NH, JS, and RA are employees of Adelphi Values.

**Availability of Data and Material** Not applicable.

**Ethics Approval** Not applicable.

**Consent to Participate** Not applicable.

**Consent for Publication** Not applicable.

**Code Availability** Not applicable.

**Author Contributions** All authors contributed to the article conceptualization and development. All authors read and approved the final version.

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